

University of Cape Town  
**University Examinations – June 2006**  
**Department of Mathematics and Applied Mathematics**  
MAM2000W, 2002S, 2004H  
**Module 2LA : Linear Algebra**  
**Solutions and mark scheme**

**Section A**

1. (a)  $\det(A) = 12$ ,  $NS(A) = \{\mathbf{0}\}$ . (2)  
(b) Yes. Since  $\det(A) \neq 0$ ,  $A$  is invertible. (1)  
(c)

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & -1 & -2 \\ -2 & 1 & 2 \end{pmatrix} \xrightarrow{R_3 \rightarrow R_3 + R_2} \begin{pmatrix} 1 & 0 & 0 \\ 2 & -1 & -2 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\text{Nullity}(B) = 1, \text{Rank}(B) = 2. \quad (2)$$

$$(d) \text{Basis for } NS(B) = \left\{ \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix} \right\}. \quad (2)$$

$$\text{Basis for } CS(B) = \left\{ \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \right\}. \quad (1)$$

$$(e) C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{pmatrix}, E_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}, E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (4)$$

$$(f) 0 = \begin{vmatrix} \lambda - 2 & 0 & 0 \\ 2 & \lambda - 4 & -2 \\ -2 & 1 & \lambda - 1 \end{vmatrix} = (\lambda - 2)[(\lambda - 4)(\lambda - 1) + 2] = (\lambda - 2)(\lambda - 2)(\lambda - 3),$$

hence the eigenvalues are  $\lambda = 2, \lambda = 3$ . (3)

(g) Eigenspace for  $\lambda = 3$  is  $W_3 = NS(3I - A) = NS(B)$ . From (d) a basis for this space is  $\left\{ \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix} \right\}$ . (1)

For  $\lambda = 2$  we have  $W_2 = NS(2I - A)$  and  $2I - A = \begin{pmatrix} 0 & 0 & 0 \\ 2 & -2 & -2 \\ -2 & 1 & 1 \end{pmatrix}$  reduces to  $\begin{pmatrix} 1 & -1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ . Hence a basis for  $W_2$  is  $\left\{ \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \right\}$ . (3)

- (h)  $A$  has only 2 linearly independent eigenvectors, not 3, and can therefore not be diagonalised. (2)

$$2. \quad (a) A_2 \quad (b) A_1 \quad (c) A_2 \quad [2,2,2]$$

$$3. \quad (a) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \ker(S) \Leftrightarrow x_2 = 0 \text{ and } x_1 + 2x_3 = 0. \text{ Hence a basis for } \ker(T) \text{ is} \\ \left\{ \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix} \right\}. \quad (2)$$

$$(b) \operatorname{ran}(T) = \left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} : a, b \in \mathbf{R} \right\}, \text{ with basis } \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\}. \quad (3)$$

$$(c) \text{ Only statement (iv) is true.} \quad (1)$$

$$(d) T \left( \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right) = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, T \left( \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right) = \begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix}, T \left( \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right) = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} \\ \text{and hence } {}_C[T]_B = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & 0 \\ -1 & -1 & 0 \\ 1 & 2 & 3 \end{pmatrix}. \quad (4)$$

$$4. \quad \langle \mathbf{v} + \mathbf{u}, \mathbf{u} + \mathbf{w} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle + \|\mathbf{u}\|^2 + \langle \mathbf{u}, \mathbf{w} \rangle = -2 + 3 + 16 - 1 = 16. \quad (2)$$

$$\|\mathbf{v} - 3\mathbf{w}\|^2 = \langle \mathbf{v} - 3\mathbf{w}, \mathbf{v} - 3\mathbf{w} \rangle = \|\mathbf{v}\|^2 - 3\langle \mathbf{v}, \mathbf{w} \rangle - 3\langle \mathbf{w}, \mathbf{v} \rangle + 9\|\mathbf{w}\|^2 = 216, \\ \text{so } \|\mathbf{v} - 3\mathbf{w}\| = 6\sqrt{16}. \quad (2)$$

### Section B

$$5. \quad (a) \text{ True.}$$

$$\begin{aligned} \lambda_1(\mathbf{v}_1 - \mathbf{v}_2) + \lambda_2(\mathbf{v}_1 + \mathbf{v}_3) + \lambda_3(\mathbf{v}_1 - 2\mathbf{v}_2 + 3\mathbf{v}_3) &= \mathbf{0} \\ \Rightarrow (\lambda_1 + \lambda_2 + \lambda_3)\mathbf{v}_1 + (-\lambda_1 - 2\lambda_3)\mathbf{v}_2 + (\lambda_2 + 3\lambda_3)\mathbf{v}_3 &= \mathbf{0} \end{aligned}$$

This gives a system of homogeneous equations in the variables  $\lambda_1, \lambda_2, \lambda_3$  with coefficient matrix  $\begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & -2 \\ 0 & 1 & 3 \end{pmatrix}$  which reduces to  $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ . From this we get  $\lambda_1 = \lambda_2 = \lambda_3 = 0$ . (4)

$$(b) \text{ False. } A_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \text{ and } A_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \text{ are both in } S, \text{ but } \det(A_1 + A_2) = 1 \neq 0, \text{ so } A_1 + A_2 \notin S. \text{ Therefore } S \text{ is not closed under addition, and therefore not a subspace.} \quad (3)$$

$$(c) \text{ False. } F \text{ is not linear since } F(\mathbf{0}) = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \neq \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}. \quad (2)$$

$$6. \quad (a) \text{ It follows from (2) that } \lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2 \cdots + \lambda_n\mathbf{v}_n \in \ker(T). \text{ Since } T \text{ is one-to-one, } \ker(T) = \{\mathbf{0}\} \text{ and therefore } \lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2 \cdots + \lambda_n\mathbf{v}_n = \mathbf{0}. \quad (2)$$

$$(b) \text{ The linear independence of } S \text{ is used to deduce (4) from (3).} \quad (1)$$

$$(c) \text{ The linearity of } T \text{ is used at (2) to deduce that } T(\lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2 \cdots + \lambda_n\mathbf{v}_n) = \lambda_1T(\mathbf{v}_1) + \lambda_2T(\mathbf{v}_2) \cdots + \lambda_nT(\mathbf{v}_n). \quad (1)$$

- (d) (i) The statement is false; it will be true if it is also known that  $\text{span}(S) = V$  (i.e. that  $S$  is a basis for  $V$ ). (1)
- (ii) The statement is false. It will be true if it is also known that  $S$  spans  $V$  and that  $T$  is surjective. (2)
- (iii) The statement is true. If  $S$  is a basis for  $V$ , then  $\dim(V) = n$ . By (i),  $C$  will be a basis for  $\text{ran}(T)$ , and since  $C$  has  $n$  elements,  $\dim(\text{ran}(T)) = n = \dim(V)$ . (2)
- (e) False. If, for example,  $T$  is the zero transformation, a linearly independent set will be mapped to  $\{\mathbf{0}\}$ , which is linearly dependent. (2)
7. (a) True. Take, for example,  $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ .
- Then  $NS(A) = CS(A) = \left\{ \lambda \begin{pmatrix} 1 \\ 0 \end{pmatrix} : \lambda \in \mathbf{R} \right\}$ . (2)
- (b) False. If  $NS(A) = CS(A)$ , then  $\text{Nullity}(A) = \text{Rank}(A) = k$  (say). But  $3 = \text{Nullity}(A) + \text{Rank}(A) = 2k$ , a contradiction. (1)
- (c) False. If  $A$  is an  $n \times n$  matrix and  $NS(A) = CS(A)$ , then for every  $\mathbf{x} \in \mathbf{R}^n$ ,  $A\mathbf{x} \in CS(A) = NS(A)$ , and so  $A(A\mathbf{x}) = \mathbf{0}$ . It follows from this that  $A^2$  is the zero matrix. Hence  $0 = \det(A^2) = (\det(A))^2$ , so  $\det(A) = 0$ . It follows from this that  $0$  is an eigenvalue of  $A$ . If  $\lambda$  is any eigenvalue of  $A$ , then there is a non-zero  $\mathbf{x} \in \mathbf{R}^n$  such that  $A\mathbf{x} = \lambda\mathbf{x}$ . Then  $\mathbf{0} = A^2(\mathbf{x}) = A(\lambda\mathbf{x}) = \lambda A\mathbf{x} = \lambda^2\mathbf{x}$ . Since  $\mathbf{x} \neq \mathbf{0}$ , we must have  $\lambda = 0$ , so  $0$  is the only eigenvalue of  $A$ . If  $A$  can be diagonalised, there is an invertible matrix  $P$  such that  $P^{-1}AP = D$ , and  $D$  will be the zero matrix in this case. But then  $A$  is the zero matrix, a contradiction. (4)